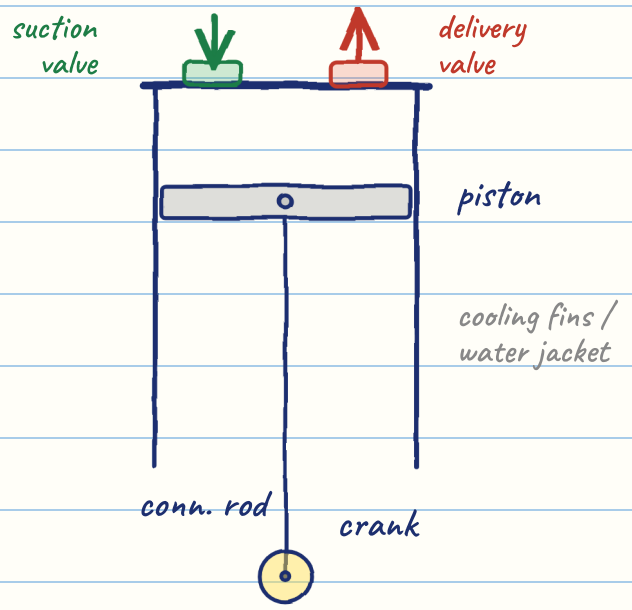
UNIT – V : AIR COMPRESSORS

5.1 Introduction & Classification

- **Compressor** : a machine that **raises the pressure of a gas (air)** by doing work on it - mechanical energy is converted into pressure energy. Uses : pneumatic tools, drilling, spray painting, air conditioning, IC-engine supercharging, gas turbines, refrigeration.
- By pressure ratio $r_p = p_2/p_1$: **Fan** ($r_p < 1.1$), **Blower** (1.1 - 4), **Compressor** (> 4).
- **(a) Positive displacement** : volume of trapped air is reduced mechanically.
  → **Reciprocating** (piston-cylinder) ; **Rotary** : Roots blower, vane (sliding vane), screw, lobe, scroll.
- **(b) Dynamic (continuous flow)** : kinetic energy given to the air is converted into pressure in a diffuser : **Centrifugal** (radial) and **Axial-flow** compressors.
- Reciprocating = high pressure, small flow ; rotary = medium ; dynamic = very large flow, moderate pressure.

5.2 Reciprocating Compressor



Suction stroke : piston ↓, suction valve opens, air enters.

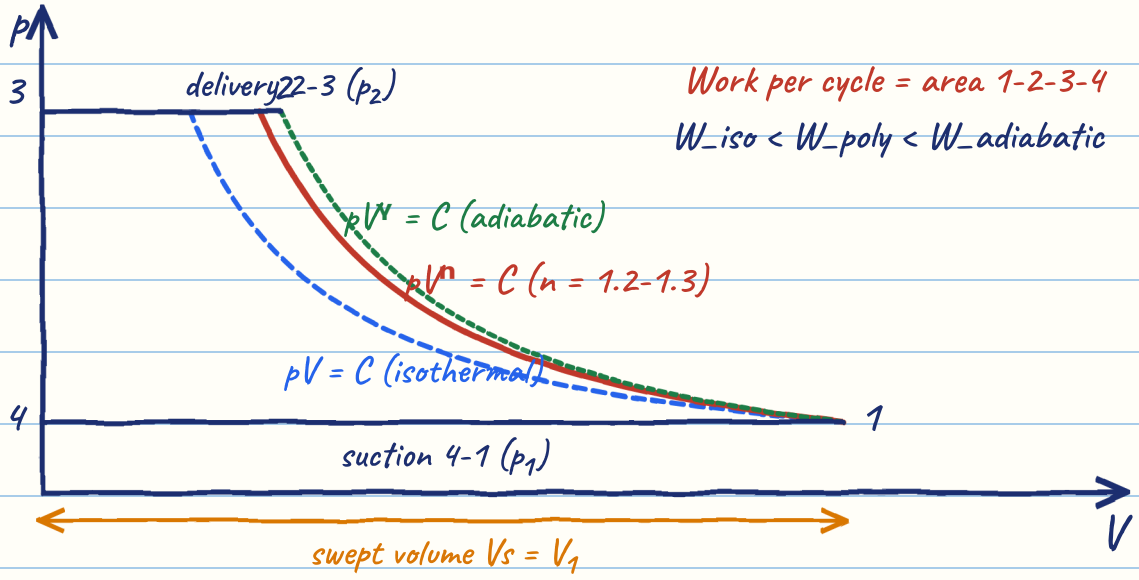
Compression : both valves shut, air compressed (p ↑, V ↓).

Delivery : delivery valve opens when p = receiver pressure.

Valves are automatic (spring-loaded plate / reed) - opened by Δp.

- **Parts** : cylinder, piston, connecting rod, crank, **suction & delivery valves** (spring-loaded, automatic), cooling jacket / fins, air receiver, filter, lubrication system, safety valve.
- May be **single acting** (one face) or **double acting** (both faces), single or multi-stage, vertical / horizontal / V / W arrangement.
- **Working** : (1) suction at p_1 with delivery valve closed, (2) compression - both valves closed, p rises, (3) **delivery** when $p = p_2$ (receiver pressure) the delivery valve opens & air is pushed out at constant p_2, (4) piston at end, clearance air remains.

5.3 Work of Compression – no clearance



- The cycle is 4-1 (suction), 1-2 (compression), 2-3 (delivery), 3-4 (pressure drop). Work per cycle = area enclosed = $\int V\,dp$.
- **Isothermal** ($pV = C$; perfect cooling):

$$W_{iso} = p_1 V_1 \ln(p_2/p_1)$$

- **Polytropic** ($pV^n = C$, $n = 1.2\text{-}1.35$):

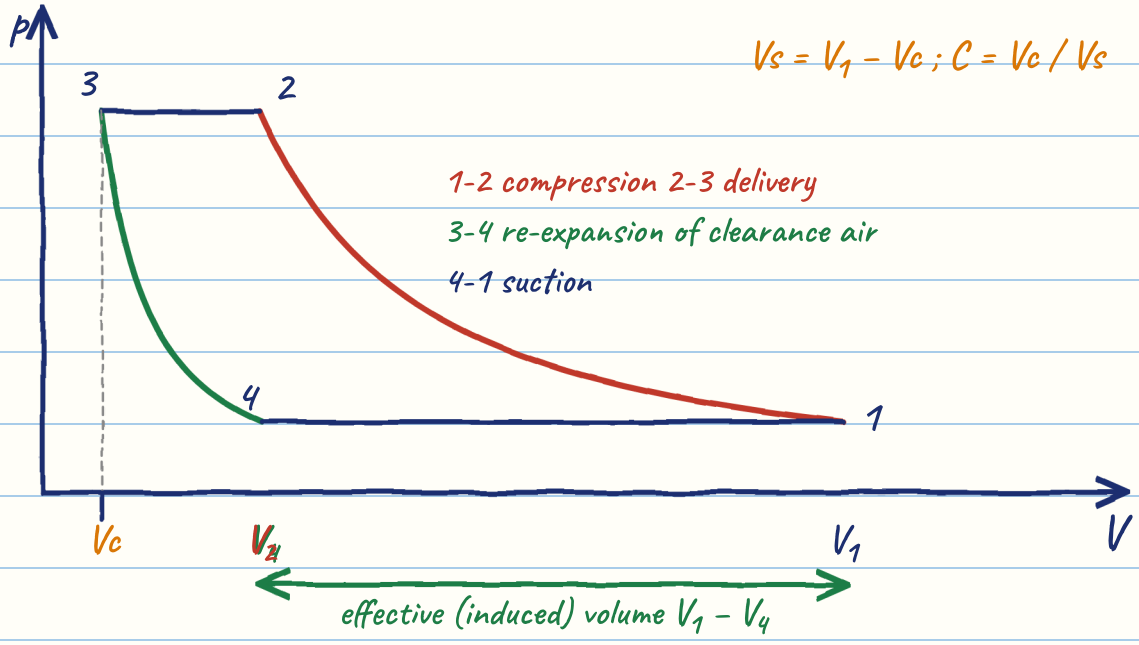
$$W = [n/(n-1)]\, p_1 V_1 [(p_2/p_1)^{(n-1)/n} - 1]$$

- **Adiabatic** ($n = \gamma = 1.4$, no cooling): same with γ in place of $n \Rightarrow$ **maximum work**.
- Delivery temperature: $T_2 = T_1 (p_2/p_1)^{(n-1)/n}$.

$W_{iso} < W_{poly} < W_{adiabatic} \Rightarrow$ cool the cylinder as much as possible (jackets, fins, intercooler) to approach isothermal.

- **Power** (single-acting, single cylinder): $P = W \times N / 60$ (W per cycle, N rpm); double acting $\Rightarrow \times 2$.
- **Isothermal efficiency** $\eta_{iso} = W_{iso} / W_{actual\ (indicated)}$ (typically 65-80 %).
- **Mechanical efficiency** $\eta_m = IP$ / shaft power (80-90 %).

5.4 Effect of Clearance Volume



$V_s = V_1 - V_c$; $C = V_c / V_s$

- A clearance volume V_c (3-10 % of V_s) is needed so the piston does not strike the head and valves can operate. After delivery, the clearance air at p_2 **re-expands** (3-4) to p_1 before the suction valve can open ⇒ part of the stroke is lost.
- **Effective (induced) volume = $V_1 - V_4$** ; swept volume $V_s = V_1 - V_c$; **clearance ratio $C = V_c / V_s$.**
- With polytropic re-expansion $V_4 = V_c (p_2/p_1)^{1/n}$:

$$\eta_v = (V_1 - V_4)/V_s = 1 + C - C\,(p_2/p_1)^{1/n}$$

- **Effects of clearance : (1) reduces the volumetric efficiency & capacity** ($\eta_v \downarrow$ with $r_p \uparrow$ and $C \uparrow$); (2) **no effect on work per kg of air** (work of compression and re-expansion cancels for same n); (3) at very high r_p, $\eta_v = 0$: $(p_2/p_1)_{max} = (1 + 1/C)^n$ ⇒ the reason for multi-staging; for $C = 0.05$, $n = 1.3 \rightarrow r_{max} = 52$.

5.5 Volumetric Efficiency & FAD

- **Volumetric efficiency** η_v = volume of free air induced / swept volume (clearance, valve losses, heating of the intake air, leakage past rings & valves, pressure drop in the suction valve lower it). Typically 60-90 %.
- **Overall (free-air) volumetric efficiency** η_{vo} = FAD per stroke / V_s (reduce to atmospheric p, T).
- **FAD (free air delivery)** = volume of air delivered reduced to atmospheric (intake) conditions : $V_{FAD} = p\,V\,T_a / (p_a\,T)$.
- **Compressor capacity** = FAD per unit time. **Indicated power** from indicator diagram.

Worked example 1

- Single-acting single-stage compressor, bore 150 mm, stroke 200 mm, N = 300 rpm, p_1 = 1 bar, p_2 = 7 bar, n = 1.3, clearance 5 % of swept volume. Find η_v, FAD (induced volume/min), IP and η_{iso}.

  → $V_s = (\pi/4)(0.15)^2(0.2) = 3.534\times10^{-3}\ m^3$

  → $\eta_v = 1 + 0.05 - 0.05\,(7)^{1/1.3} = 1.05 - 0.05(4.467) =$ **0.827**

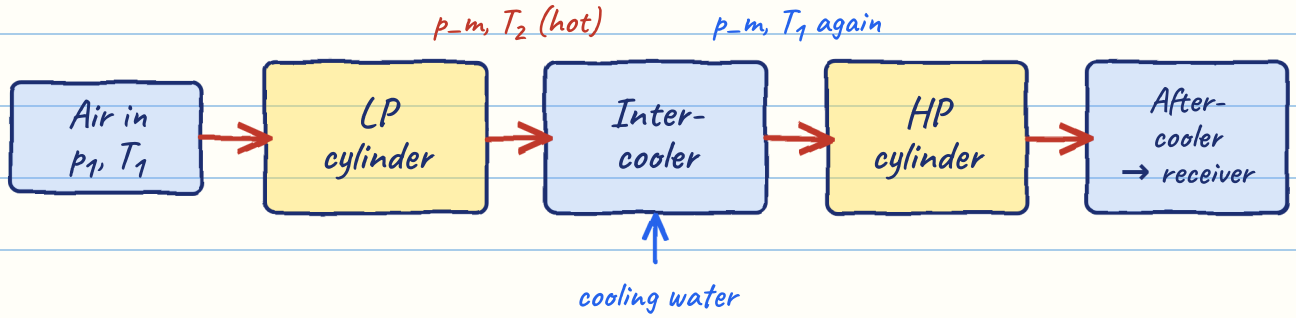
  → Induced volume per cycle = $0.827 \times 3.534\times10^{-3} = 2.921\times10^{-3}\ m^3$; per minute = **0.876 m^3/min**

  → $W = (1.3/0.3) \times 10^5 \times 2.921\times10^{-3} \times [7^{0.3/1.3} - 1] = 4.333 \times 292.1 \times 0.567 =$ **717.6 J/cycle**

  → $IP = 717.6 \times 300/60 =$ **3.59 kW**; $W_{iso} = 10^5 \times 2.921\times10^{-3} \times \ln 7 = 568.5$ J $\Rightarrow \eta_{iso} = 568.5/717.6 =$ **79.2 %**

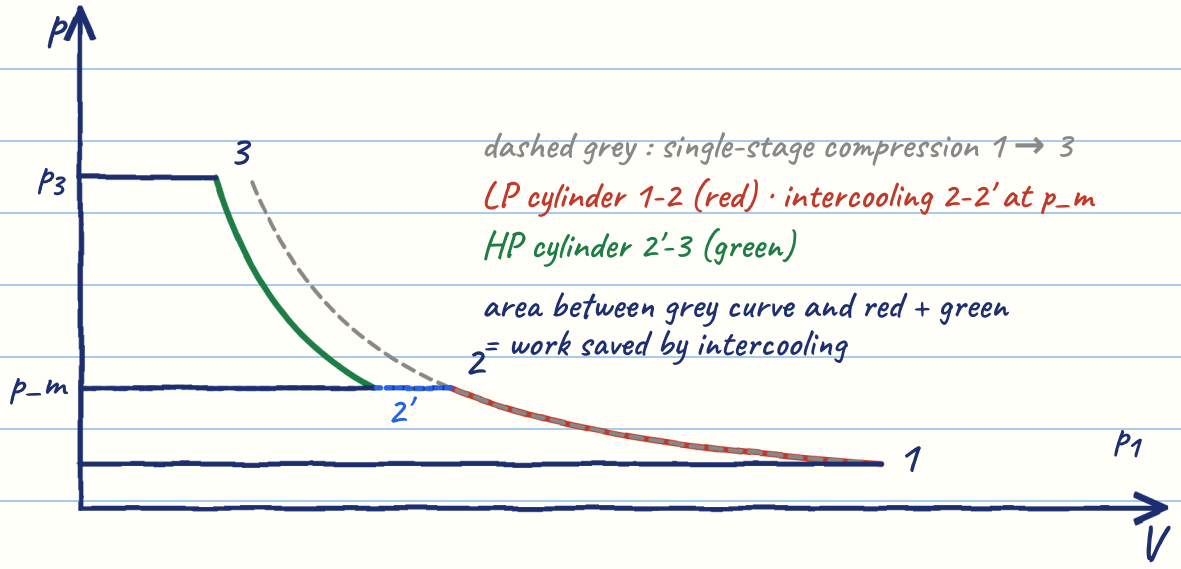
5.6 Multi-stage Compression – Why ?

- For high pressures (> 8 bar) a single stage is bad because : (1) **delivery temperature becomes very high** ($T_2 = T_1\, r_p^{(n-1)/n} \approx 300$ °C at $r_p = 16$) → lubricating oil flash point (≈ 200 °C) & valve problems, (2) **η_v falls** (clearance effect), (3) **more work** (poor cooling), (4) large cylinder size & heavy construction for high pressure / strong flywheel, (5) uneven torque.
- **Multi-stage** : compress in 2 or more cylinders with **intercoolers** between them. Advantages : saving of work, better η_v, lower delivery temperature, lighter cylinders (smaller HP cylinder), better balancing & more uniform torque. Disadvantage: more cost, space.



Perfect intercooling : air leaves intercooler at the original temperature T_1 ($T_3' = T_1$).
Minimum work for $p_m = \sqrt{(p_1\, p_3)}$ → equal pressure ratio in each stage.

5.7 Two-stage Compressor with Intercooling



- LP cylinder compresses 1→2 ($p_1 \to p_m$). The air passes through an **intercooler** at constant pressure p_m cooled 2→2' back to the initial temperature (**perfect intercooling**, $T_{2'} = T_1$, hence $p_m V_{2'} = p_1 V_1$). HP cylinder then compresses 2'→3 ($p_m \to p_3$).
- **Total work** (perfect intercooling, same n) :

$$W = [n/(n-1)]\, p_1 V_1 \left[(p_m/p_1)^{(n-1)/n} + (p_3/p_m)^{(n-1)/n} - 2 \right]$$

- **Optimum intermediate pressure** ($dW/dp_m = 0$) :

$$p_m = \sqrt{(p_1 p_3)} \Rightarrow p_m/p_1 = p_3/p_m$$

- Minimum work : $W_{min} = [2n/(n-1)]\, p_1 V_1 \left[(p_3/p_1)^{(n-1)/2n} - 1 \right]$
- For **z stages** : pressure ratio per stage $r = (p_{final}/p_1)^{1/z}$; $W_{min} = z\, n/(n-1)\, p_1 V_1 \left(r^{(n-1)/n} - 1\right)$. Work in each stage is equal ⇒ **equal work per stage**.
- **Heat rejected in the intercooler** per kg = $c_p (T_2 - T_1) = c_p T_1 \left(r^{(n-1)/n} - 1\right)$. Number of stages is chosen so that the delivery temperature per stage < 150-180 °C (usually 2 stages up to 30 bar, 3 up to 100 bar).

- **Interstage area** : cylinder volume ratio (LP : HP) = r (for equal ratios) → HP cylinder diameter smaller.

5.8 Effect of Pressure Drop & Imperfect Intercooling

- **Imperfect intercooling** : $T_{2'} > T_1 \Rightarrow p_m V_{2'} > p_1 V_1 \Rightarrow$ HP cylinder handles more volume ⇒ **more HP work**, saving is reduced; p_m shifts slightly.
- **Pressure drop in the intercooler / pipes / valves** : the HP cylinder draws air at $p_m' <$ LP delivery pressure $p_m \Rightarrow$ the HP suction line lies lower & LP delivery higher ⇒ **extra loop area (loss of work)**, lower η_v, higher power per m^3 of FAD. Keep intercooler $\Delta p < 0.1$-0.2 bar by large flow area.
- Also leakage in valves & rings, heating of suction air, and delivery valve resistance - increase the indicated work & reduce capacity.

Worked example 2 (two-stage)

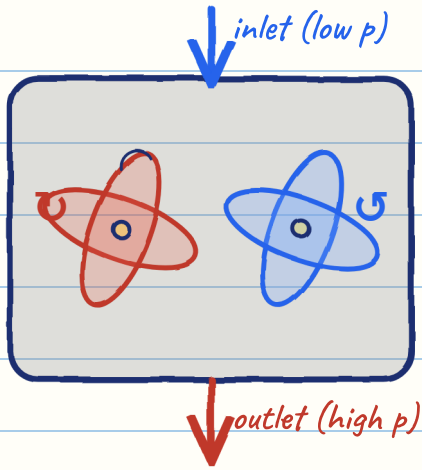
- A two-stage compressor delivers FAD 3 m^3/min from 1 bar, 27 °C to 16 bar, n = 1.3, perfect intercooling, optimum p_m. Find p_m, power & saving over single stage.
  - → $p_m = \sqrt{(1 \times 16)} =$ **4 bar** ; $p_1 V_1 = 10^5 \times 3/60 = 5000$ W
  - → $W_2 = (2 \times 1.3/0.3) \times 5000 \times [16^{0.3/2.6} - 1] = 8.667 \times 5000 \times 0.377 =$ **16.34 kW**
  - → Single stage : $W_1 = 4.333 \times 5000 \times [16^{0.3/1.3} - 1] = 21667 \times 0.896 =$ **19.42 kW** ⇒ saving ≈ **3.08 kW (≈ 16 %)**
  - → Delivery temp.: single stage 296 °C vs 140 °C per stage in two-stage ✓

5.9 Problems Related to Reciprocating Compressors

Problem	Cause / remedy
Valve leakage / failure	dirt, broken springs, carbon ⇒ filter, clean & replace valves
High discharge temperature	poor cooling, fouled intercooler ⇒ clean, more stages
Carbon deposits, fire / explosion	oil oxidation at high T near valves ⇒ right oil, cooling
Water carry-over	moist air condensed in intercooler ⇒ drain traps, aftercooler
Pulsating flow & vibration	intermittent delivery ⇒ air receiver, balancing, dampers
Ring & liner wear, oil carry-over	dust & poor lubrication ⇒ filter, oil separator
Knocking	liquid/water slugging, loose parts, excess clearance
Overheating of cylinders	low coolant flow ⇒ check pump, fins, jackets
Capacity control	unloaders, clearance pockets, speed variation, start-stop

- Safety valve on receiver & intercooler; pressure gauge; drain cocks; relief valves between stages; fusible plug for fire.

5.10 Rotary Compressors – Roots Blower



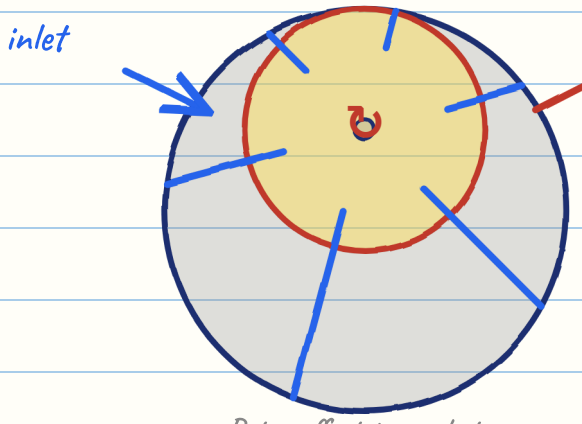
Two 2-lobe (figure-8) rotors, synchronised by timing gears, turn in opposite directions.

Air trapped between lobe & casing is carried to the delivery side - no internal compression (volume constant).

Pressure rises when delivery side back-flows.

- **Working principle** : two figure-of-eight (**two or three lobe**) rotors synchronised by timing gears rotate in opposite directions inside a casing with small (≈ 0.1 mm) clearances. Air is trapped between lobe & casing & **carried round** to the delivery side; there is **no internal compression** - pressure rises when the delivery is closed (back-flow compresses it). Delivery is pulsating.
- Features : simple, oil-free air (no contact), moderate pressure ratio (≤ 2, up to 0.5-1 bar), efficiency 50-70 %; used as supercharger, scavenging blower for 2-stroke diesels, vacuum pumps, pneumatic conveying, ventilation.
- Capacity ∝ speed (nearly constant volume machine).

5.11 Vane Type Blower (Sliding-vane compressor)



Rotor offset towards top

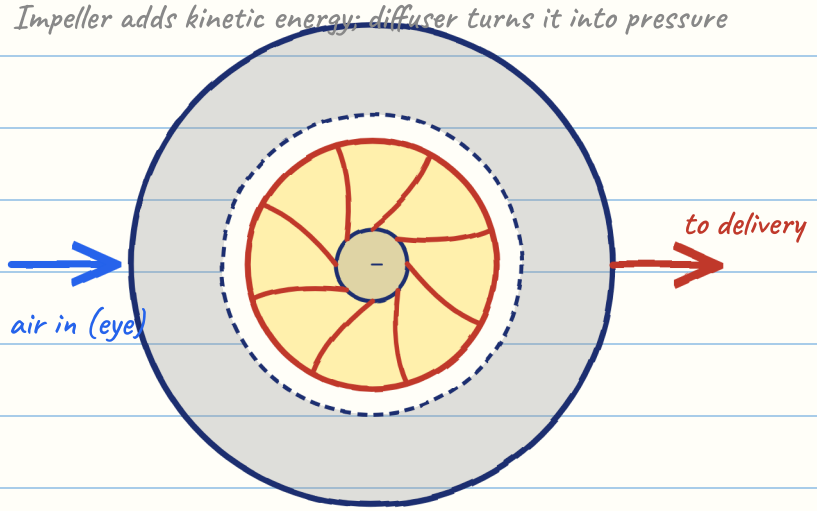
Eccentric rotor with sliding vanes in radial slots; centrifugal force presses vanes on the casing.

Trapped volume between vanes first increases (suction) then decreases (internal compression) → delivery.

- **Working** : a cylindrical rotor is mounted eccentrically in a cylindrical casing; **vanes** slide in radial slots & are thrown against the casing wall by centrifugal force. As the rotor turns the volume between consecutive vanes **increases at the suction port (suction)** and then **decreases (internal compression)** before the delivery port.
- Features : compact, smooth flow, oil-lubricated (sealed by oil), p-ratio up to 4-5 in single stage (8 for two-stage); vane wear & frictional heating (oil cooled). Used in refrigeration & air-conditioning compressors, vacuum pumps, small air supply, car power-steering, fuel pumps.

5.12 Centrifugal Compressor

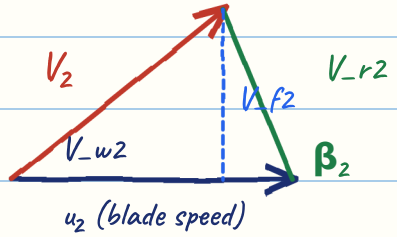
Impeller adds kinetic energy; diffuser turns it into pressure



Impeller (red) + vaneless / vaned diffuser (dashed) + volute casing

$W = u_2 V_{w2}$ (J/kg)

Outlet velocity triangle



- **Parts** : inlet casing, **impeller** (rotating blades, 3000-30000+ rpm), **diffuser** (vaneless / vaned) and **volute** (scroll) casing. Air enters axially at the eye, **centrifugal action** throws it outward ⇒ both **pressure & velocity** rise in the impeller; diffuser converts high velocity (kinetic energy) into pressure; volute collects & delivers.
- **Euler work** (per kg) : $W = u_2 V_{w2} - u_1 V_{w1}$; for axial entry $V_{w1} = 0$:

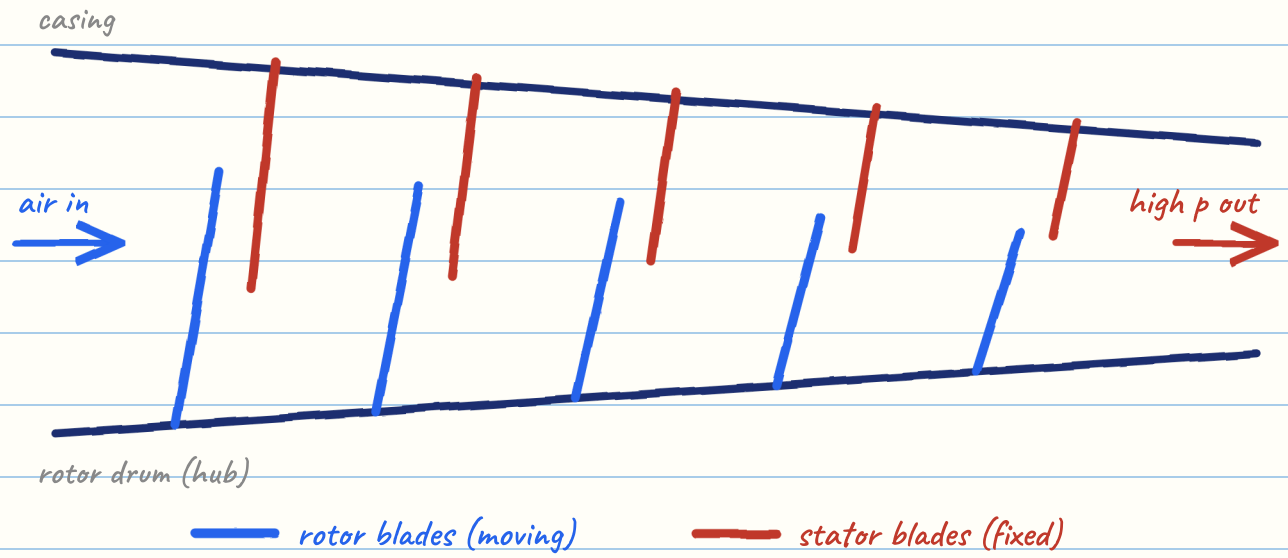
$$W = \sigma \psi u_2^2 = c_p (T_{02} - T_{01})$$ (σ = slip factor, ψ = work-input factor ≈ 1)

- **Slip** : the actual V_{w2} < ideal because the air lags behind the blade ⇒ slip factor $\sigma = V_{w2}/V_{w2}' \approx 0.85\text{-}0.9$.
- **Pressure ratio** per stage 3 - 5 (up to 8-10 with titanium impeller); isentropic efficiency 75-85 %.

$$p_{02}/p_{01} = [1 + \eta_c \psi \sigma u_2^2 / (c_p T_{01})]^{\gamma/(\gamma-1)}$$

- **Limits** : **surge** (flow reversal at low flow ⇒ violent pulsation), **choking** (Mach 1 at the inducer / diffuser throat at high flow); the **operating range** lies between surge line & choke line.
- Uses : gas-turbine & turbocharger compressors (small), large process / blast-furnace air, refrigeration (chillers), air-conditioning. Suited for large flow rates, compact, oil-free, steady delivery.

5.13 Axial Flow Compressor



- **Principle** : air flows **parallel to the axis** through alternating rows of **rotor blades** (give kinetic energy / raise velocity & some pressure) & **stator blades** (diffuse: convert velocity into pressure and guide air to the next rotor). One rotor row + one stator row = **one stage**; annulus area decreases to keep axial velocity constant as density rises.
- Typical stage pressure ratio only **1.1 - 1.4** (ΔT 15-30 K) ⇒ many stages (8 to 20) are used for an overall ratio 10-40.

- **Stage work** : $W = u\,(V_{w2} - V_{w1}) = c_p \Delta T_{0s}$; degree of reaction $R \approx 0.5$ (symmetrical blading); **de Haller number** ($V_2/V_1 > 0.72$) limits diffusion to avoid blade stall.
- **Stall & surge** : rotating stall (cells of separation) and surge limit the stable operating range. Isentropic efficiency 85-92 %.
- Uses : **jet engines & gas turbines**, power-plant, wind tunnels, blowers.

5.14 Comparison of Compressors

Point	Reciprocating	Roots	Vane	Centrifugal	Axial
Type	positive	positive	positive	dynamic	dynamic
Flow	small	medium	small-med.	large	very large
Pressure ratio	very high	≤ 2	4 - 8	3 - 10	10 - 40 (multi-stage)
Delivery	pulsating	pulsating	smooth	steady	steady
Efficiency	70-85 %	50-70 %	60-75 %	75-85 %	85-92 %
Speed	low	medium	medium	very high	very high
Size / cost	big, heavy	small	small	compact	compact, costly

Important questions (exam)

- Derive work of single-stage reciprocating compressor : isothermal & polytropic.
- Derive volumetric efficiency in terms of clearance ratio & pressure ratio.
- Derive the condition for minimum work in two-stage compressor with intercooler; show $p_m = \sqrt{(p_1 p_2)}$.

- Effect of pressure drop & imperfect intercooling in multistage compressors.
- Working of Roots blower, vane blower, centrifugal & axial-flow compressors (neat sketches).